

Name: _____ Roll No. _____

Institute of Engineering & Technology, DAVV, Indore

B.E. First Year, Class Test-III, (IT- A and B)

IACRC2-Chemistry and Environmental Science

Time: 70

Max Marks: 20

Note: Attempt ALL questions.

Q.1 Determine correct structure for two compounds with same molecular formula $C_4H_8O_2$ and having following observations in their NMR spectrum, respectively-

Compound A: Doublet (1.26 ppm), Septet (2.30 ppm) and Singlet (1.08 ppm) with Peak height ratio of **6:1:1**

Compound B: Triplet (1.26 ppm), Singlet (2.05 ppm), Quartet (4.12 ppm) with Peak height ratio of **3:3:2**

Q.2 What is meant by the term chemical shift? How an aldehydic proton absorbs in 1H NMR spectrum of a compound?

Q.3 Discuss flash point and fire point with their importance and explain construction of apparatus (any one) and process of its determination.

OR

Discuss construction of Redwood viscometer and its uses in the determination of viscosity-index. An oil sample under test has the same viscosity as that of the standard Gulf oil and Pennsylvanian oil at $210^\circ F$. Their viscosities at $100^\circ F$ are 60, 760 and 420 s, respectively. Calculate the viscosity index of the oil sample.

Q.4 Write a note on structure of atmosphere and atmospheric laps rate.

OR

Explain functions and classification of lubricants.
